

Exercise 15.2

line Integral

1st Formula:

$$\int_c f(x, y) ds = \int_a^b f(x(t), y(t)) \|r'(t)\| dt$$

where $r(t) = x(t)i + y(t)j$; $a \leq t \leq b$

Example 1:

Using the given parameterization evaluate the line integral $\int_c (1 + xy^2) ds$

(a) $C: r(t) = ti + 2tj$ ($0 \leq t \leq 1$)

$$x(t) = t$$

$$y(t) = 2t$$

$$r'(t) = i + 2j$$

$$\begin{aligned} \|r'(t)\| &= \sqrt{1^2 + 2^2} \\ &= \sqrt{5} \end{aligned}$$

$$\int (1 + xy^2) ds = \int_0^1 (1 + (t)(2t)^2) \sqrt{5} dt$$

$$= \sqrt{5} \int_0^1 (1 + 4t^3) dt$$

$$= \sqrt{5} \left[t + \frac{4t^4}{4} \right]_0^1$$

$$= \sqrt{5} [1 + 1^4 - 0]$$

$$= 2\sqrt{5}$$

$$(b) \quad C: r(t) = (1-t)i + (2-2t)j \quad (0 \leq t \leq 1)$$

$$x(t) = 1-t$$

$$y(t) = 2-2t$$

$$r'(t) = -i - 2j$$

$$\begin{aligned} |r'(t)| &= \sqrt{(-1)^2 + (-2)^2} \\ &= \sqrt{1+4} \\ &= \sqrt{5} \end{aligned}$$

$$\int_C (1+xy^2) ds = \int_0^1 (1+(1-t)(2-2t)^2) \sqrt{5} dt$$

$$= \sqrt{5} \int_0^1 (1+(1-t)4(1-t)^2) dt$$

$$= \sqrt{5} \int_0^1 1+4(1-t)^3 dt$$

$$= \sqrt{5} \int_0^1 1-4(-(1-t)^3) dt$$

$$= \sqrt{5} \left[t - \frac{4(1-t)^4}{4} \right]_0^1$$

$$= \sqrt{5} [1 - 0 - 0 + 1]$$

$$= 2\sqrt{5}$$

2nd formula:-

$$\int_C f(x,y) ds = \int_a^b f(x(t), y(t)) \left(\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \right) dt$$

Example 2:

Evaluate the line integral

$$\int_C (xy + z^3) ds \text{ from } (1, 0, 0) \text{ to } (-1, 0, \pi)$$

along

$$x = \cos t, y = \sin t, z = t \quad (0 \leq t \leq \pi)$$

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t, \quad \frac{dz}{dt} = 1$$

$$\begin{aligned} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} &= \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2} \\ &= \sqrt{\sin^2 t + \cos^2 t + 1} \\ &= \sqrt{1 + 1} \\ &= \sqrt{2} \end{aligned}$$

$$\begin{aligned} \int_C (xy + z^3) ds &= \\ \int_0^\pi ((\cos t \sin t) + t^3) \sqrt{2} dt \end{aligned}$$

$$= \sqrt{2} \left(\frac{\sin^2 t}{2} + \frac{t^4}{4} \right) \Big|_0^\pi$$

$$= \sqrt{2} \left(\frac{\sin^2 \pi}{2} + \frac{\pi^4}{4} \right)$$

$$= \sqrt{2} \left(\frac{\pi^4}{4} \right)$$

Q:19 $\int_c \frac{1}{1+x} ds$

$c: r(t) = ti + \frac{2}{3} t^{3/2} j \quad (0 \leq t \leq 3)$

$x(t) = t$

$y(t) = \frac{2}{3} t^{3/2}$

$r'(t) = i + t^{1/2} j$

$|r'(t)| = \sqrt{(1)^2 + (t)^{1/2 \times 2}}$
 $= \sqrt{1+t}$

$\int_c \frac{1}{1+x} ds = \int_0^3 \frac{1}{1+t} (\sqrt{1+t}) dt$

$= \int_0^3 (1+t)^{-1/2} dt$

$= \left[\frac{(1+t)^{1/2}}{1/2} \right]_0^3$

$= 2 \left[(1+3)^{1/2} - (1+0)^{1/2} \right]$

$= 2(2-1)$

$= 2$

Example 7

Evaluate

$\int_c (3x^2 + y^2) dx + 2xy dy$

along the circular arc C given by $x = \cos t$
 $y = \sin t$, $0 \leq t \leq \frac{\pi}{2}$

$$x = \cos t$$

$$y = \sin t$$

$$dx = -\sin t \, dt$$

$$dy = \cos t \, dt$$

$$= \int_0^{\pi/2} (3 \cos^2 t + \sin^2 t)(-\sin t \, dt) + 2 \cos t \sin t (\cos t \, dt)$$

$$= \int_0^{\pi/2} 3 \cos^2 t \sin t - \sin^3 t + 2 \cos^2 t \sin t \, dt$$

$$= \int_0^{\pi/2} (-\cos^2 t \sin t - \sin^3 t) \, dt$$

$$= \int_0^{\pi/2} -\sin t (\cos^2 t + \sin^2 t) \, dt$$

$$= \int_0^{\pi/2} -\sin t \, dt$$

$$= (\cos t)_0^{\pi/2} \Rightarrow \cos\left(\frac{\pi}{2}\right) - \cos(0)$$

$$= 0 - 1 \Rightarrow -1$$

$$Q:23 \quad \int_C (x + 2y \, dx + (x - y) \, dy)$$

$$C: x = 2 \cos t, \, y = 4 \sin t \quad (0 \leq t \leq \pi/4)$$

$$dx = -2 \sin t \, dt, \, dy = 4 \cos t \, dt$$

$$= \int_0^{\pi/4} [(2\cos t + 8\sin t)(-2\sin t dt) + (2\cos t - 4\sin t)(4\cos t) dt]$$

$$= \int_0^{\pi/4} [-4\cos t \sin t - 16\sin^2 t + 8\cos^2 t - 16\sin t \cos t] dt$$

$$= \int_0^{\pi/4} \left(20\cos t(-\sin t) - 16\left(\frac{1 - \cos 2t}{2}\right) + 8\left(\frac{1 + \cos 2t}{2}\right) \right) dt$$

$$= \left[\cancel{20} \frac{\cos^2 t}{2} - 8\left(t - \frac{\sin^2 t}{2}\right) + 4\left(t + \frac{\sin 2t}{2}\right) \right]_0^{\pi/4}$$

$$= [10\cos^2 t - 8 + 4\sin 2t + 4 + 2\sin 2t]_0^{\pi/4}$$

$$= [10\cos^2 t - 4t + 6\sin 2t]_0^{\pi/4}$$

$$= \left[10\cos^2 \frac{\pi}{4} - 4\frac{\pi}{4} + 6\sin \cancel{2}\left(\frac{\pi}{\cancel{4}}\right) - (10\cos 0 - \right.$$

$$\left. 0 - 6\sin 0) \right]$$

$$= \cancel{10}\left(\frac{1}{2}\right) - \pi + 6 - 10$$

$$= 1 - \pi$$